

PA2 - Matrix Transformations and Perspective

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WebGL 1.0 and glMatrix 2.8.1. Evidence from the in-app browser and native Edge. Independent Chrome/Firefox verification remains pending.

E1 - Order of transformations

Prediction: the solid uses $M = R_x(2\pi t/14) \cdot T(0,2.5,0) \cdot R_y(2t) \cdot S(s)$, where $s = 0.65 + 0.15 \sin(2\pi t/3)$. Swapping the orbit rotation and translation should remove the orbit. Swapping translation and self-spin should have no effect for this particular variant.

Result: the baseline centre at $t=3.5$ is approximately $(0,0,2.5)$. The first swap fixes it at $(0,2.5,0)$. The second swap matches the baseline; numerical comparison over 4,201 poses from 0 to 42 s found zero matrix difference. Evidence: baseline.png, swap-orbit-translation.png, swap-translation-self.png and console-output.txt in experiments/.

Explanation: glMatrix right-multiplies, so the last operation acts first on the vertex. $R_x \cdot T$ rotates the translated centre around the origin, while $T \cdot R_x$ does not. Translation parallel to y commutes with rotation about y, which explains the unchanged second swap. This is a special case, not a general claim that transformations commute.

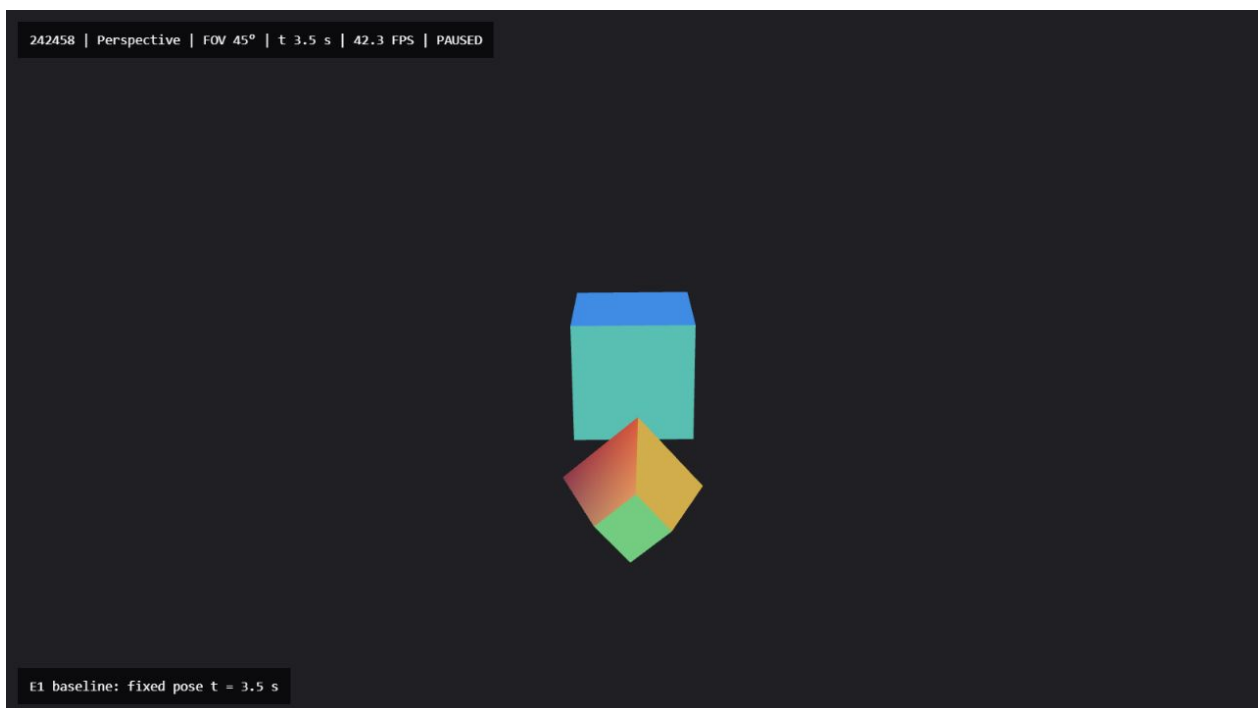


Figure 1. E1 baseline, paused at $t=3.5$ s.

E1 - Evidence of both swaps

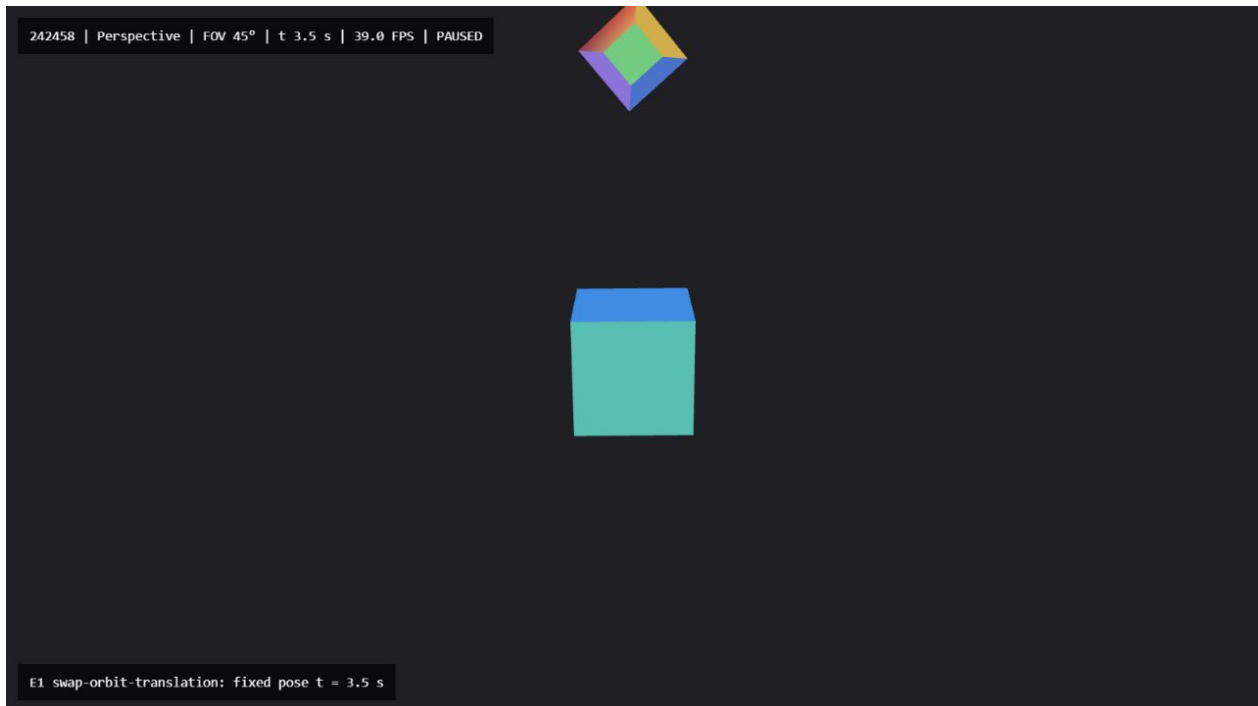


Figure 2. Orbit/translation swapped: the centre stays at (0,2.5,0).

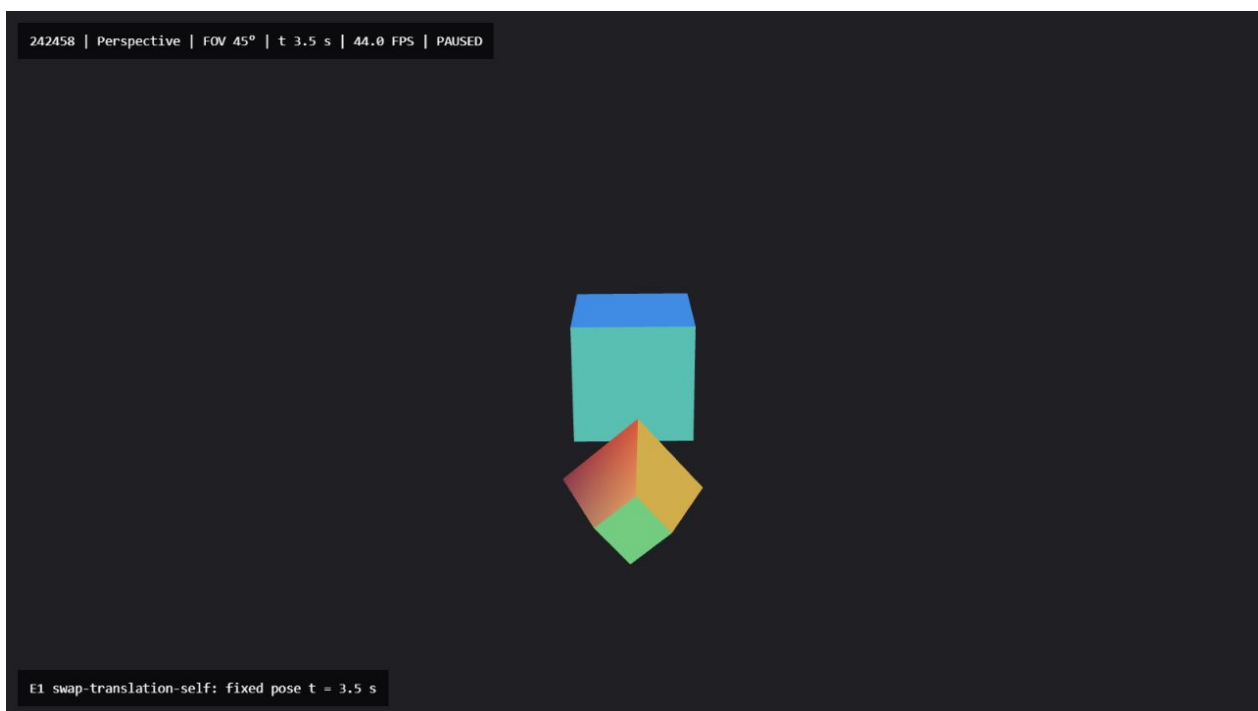


Figure 3. Translation/self-spin swapped: this y-axis variant matches the baseline.

E2 - Three statements in Chapter 4

Prediction: the axis argument, clip-volume rules and homogeneous-coordinate calculation should distinguish the three statements.

Result: an argument $[0,1,0]$ specifies the y-axis, despite the chapter's X-axis comment. Frustum clipping differs from face culling. Translating by $(2,3,4)$ leaves $[1,0,0,0]$ unchanged but maps $[1,0,0,1]$ to $[3,3,4,1]$. Evidence: homogeneous-w.png and console-output.txt.

Explanation: clipping cuts primitives against the viewing volume, potentially creating intersections; face culling rejects faces based on orientation. Homogeneous w weights translation: $w=0$ represents a direction, while $w=1$ represents a point. After perspective projection, clip xyz is divided by clip w. This use of w is distinct from a quaternion's fourth component.

```
242458 | E2 homogeneous coordinates | Console-output evidence

Axis [0,1,0], 90 degrees:
[1,0,0,1] -> [0, 0, -1, 1]

Translation: [2,3,4]
[1,0,0,0] -> [1, 0, 0, 0]
[1,0,0,1] -> [3, 3, 4, 1]
```

Figure 4. E2 translation of a direction and a point, with console results displayed.

E3 - Perspective versus orthographic

Prediction: at identical paused $t=0$, the nearer cube edge should appear longer in perspective; parallel equal edges should match in orthographic projection.

Result: at 1280×720 and FOV 45° , near/far projected edge lengths are 116.65/104.41 CSS pixels, ratio 1.1172, in perspective; they are 110.11/110.11 pixels, ratio 1.0000, in orthographic projection. Evidence: projection-perspective.png, projection-orthographic.png and E3-E4-output.txt. Measurements use endpoint projection through the actual P·V·M matrices; the hidden far edge has a cyan guide. These are not independent raster-ruler measurements.

Explanation: perspective divides by depth-dependent clip w . Orthographic w remains 1. The PA1 depth-offset trick resembles parallel projection, closer to orthographic than perspective, because depth adds a fixed screen offset without distance-dependent shrinking; it is an oblique approximation.

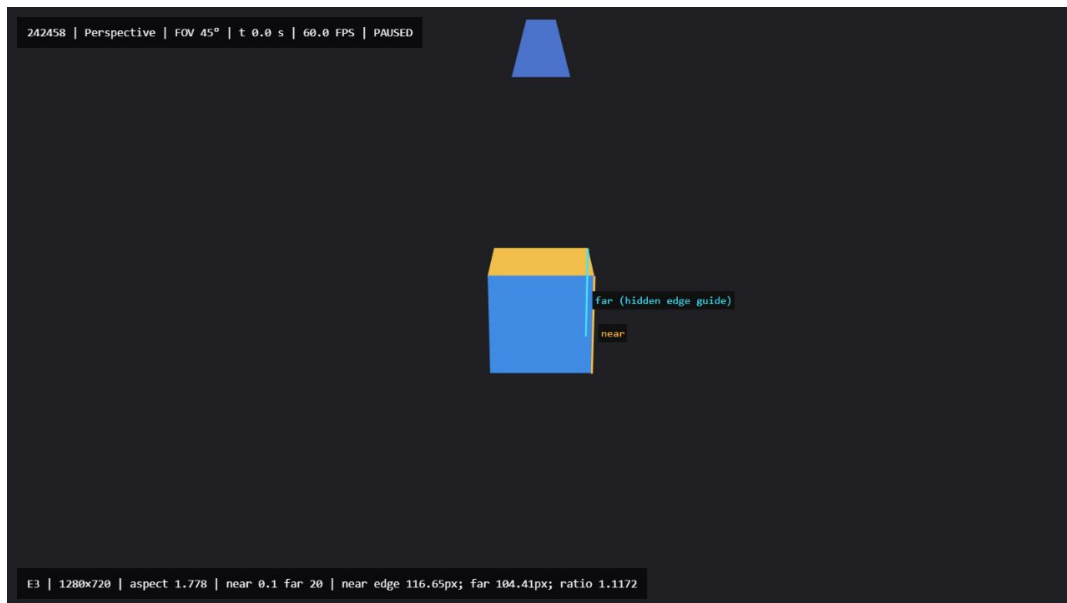


Figure 5. E3 perspective, same paused $t=0$ pose.

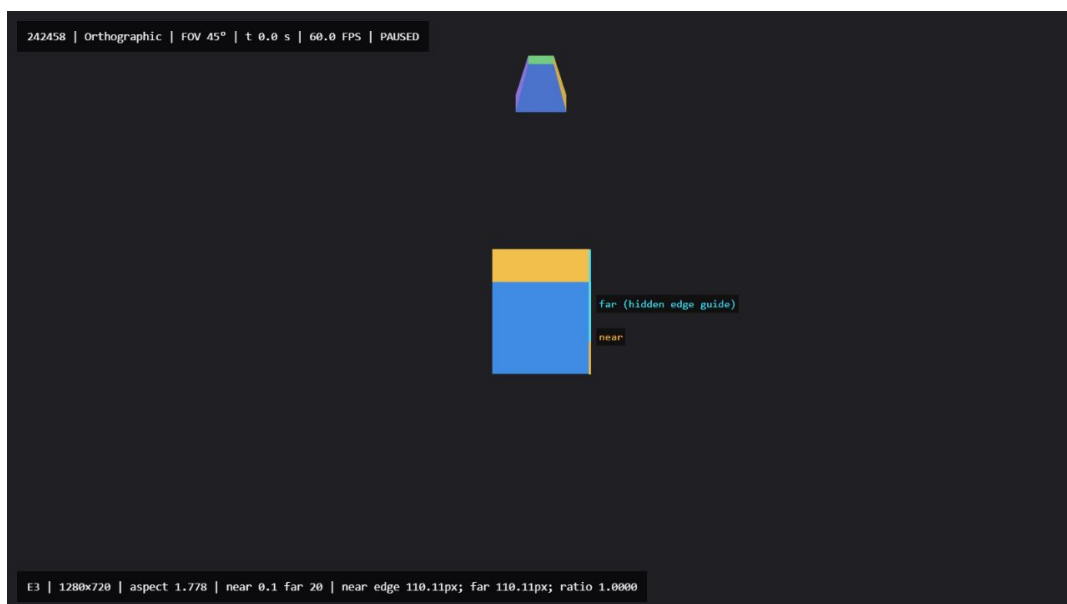


Figure 6. E3 orthographic, same paused $t=0$ pose.

E4 - Near, far and aspect

Prediction: near=0.1 and far=20 should contain the scene; near=5.2 should cut the orbiting solid. Hard-coding aspect=1 should stretch a wide viewport.

Result: the near-cut screenshot shows intersected faces at $t=3.5$, while the reference retains them. Aspect=1 is correct at 720×720 but stretches the 1440×720 view horizontally compared with aspect=2. Evidence: near-reference.png, near-cut.png, aspect-square.png, aspect-wide-one.png and aspect-wide-correct.png.

Explanation: the eye is $\sqrt{55.25} \approx 7.433$ units from the origin. All vertices lie within radius $2.5 + 0.8\sqrt{3}/2 \approx 3.193$, giving conservative eye-distance and positive view-depth bounds of approximately 4.24-10.63. Thus 0.1 and 20 bracket every pose. Numerical sampling gives tighter depths around 4.63-10.23, but sampling alone is not a continuous proof. Clipping uses view-axis depth. At width/height=2, aspect=1 doubles horizontal dimensions.

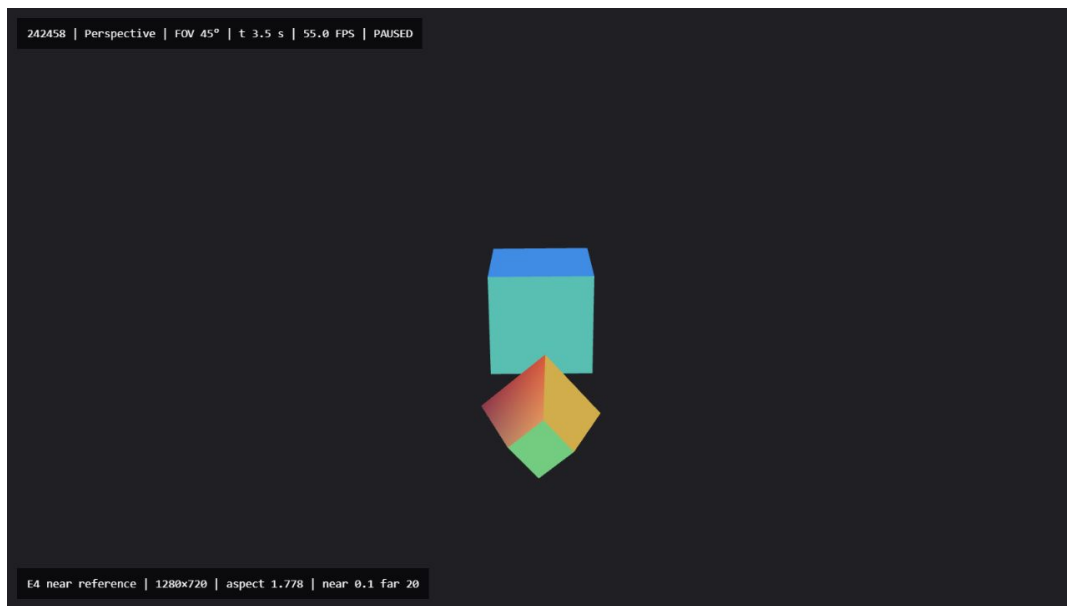


Figure 7. E4 reference near=0.1 at $t=3.5$.

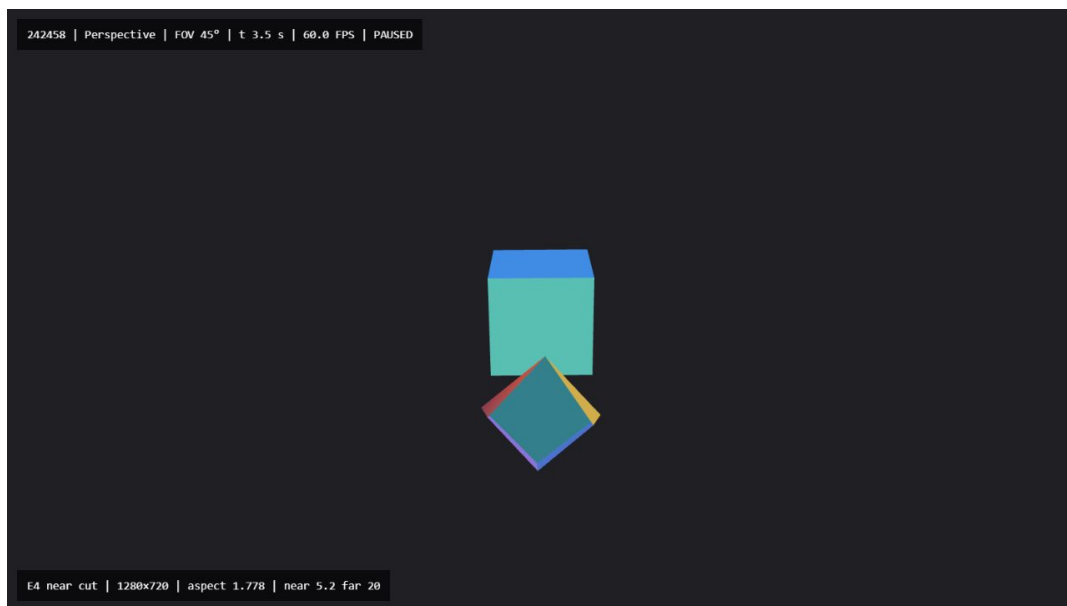


Figure 8. E4 near=5.2 intersects the frustum at the same pose.

E4 - Aspect comparison

Square and wide-window experiment copies use the same paused $t=0$ pose. The labels show viewport dimensions and the projection aspect.



Figure 9. 720x720, aspect=1.

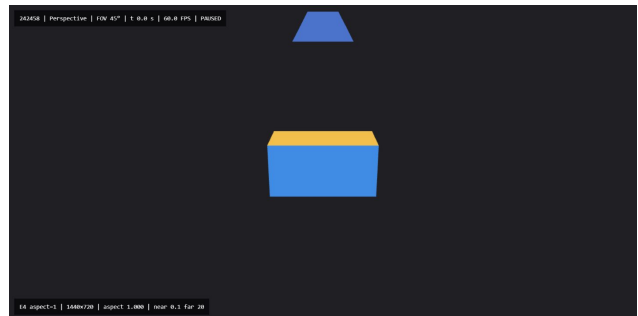


Figure 10. 1440x720, aspect=1: horizontal stretching.

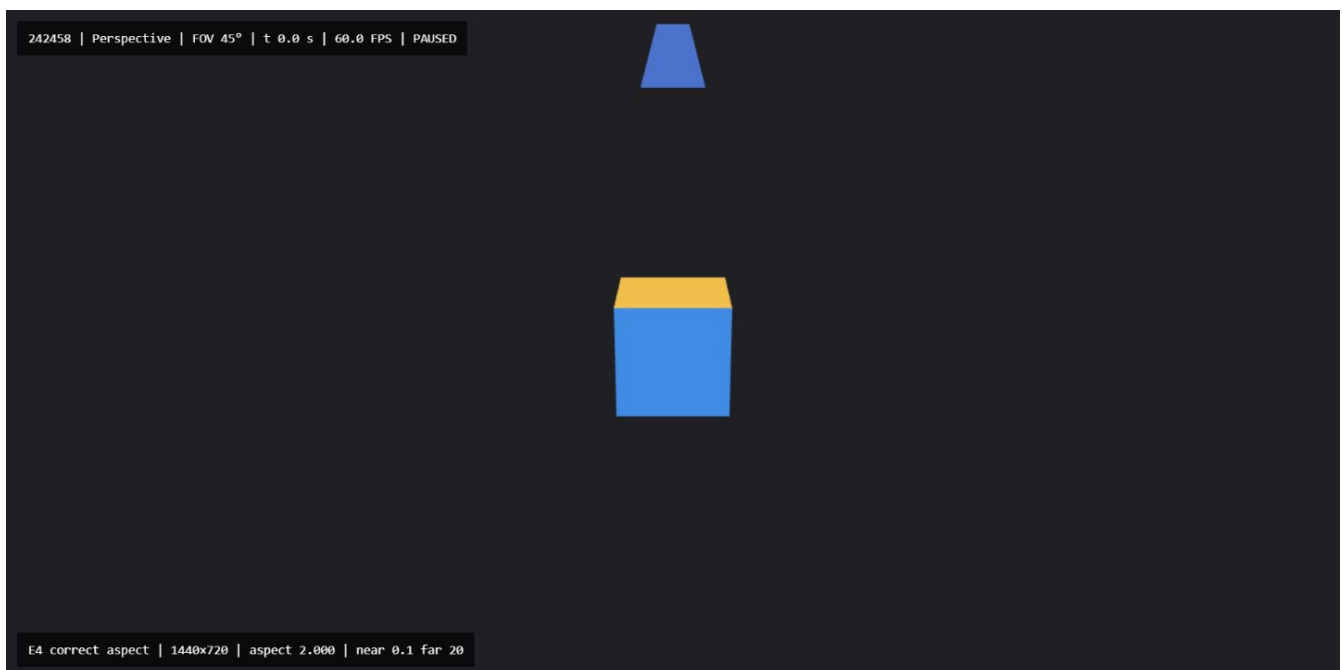


Figure 11. 1440x720, aspect=2: shape preserved.

E5 - Frame-rate independence

Prediction: accumulating dt should preserve the 14-second orbit while gaps remain below the 0.1-second clamp. Without the clamp, returning from a hidden tab should advance the pose abruptly.

Result: the baseline five-second sample contains 293 intervals, mean 0.0170648464 s, maximum 0.0999 s, and an orbit of 14.0167 s. In native Edge with DevTools CPU 4× enabled, 291 intervals over 5.000002 s give mean 0.0171821375 s and maximum 0.05 s; the orbit takes 14.000202 s. After a timed tab switch without the clamp, actual hiding lasts 5.1263 s; the first resumed dt is 5.1334 s, advancing t from 88.6986 to 93.8320. Evidence: timing-baseline.png, timing-cpu.png, timing-cpu4-results.png, timing-unclamped-return5.png and timing-unclamped-console.png.

Explanation: each orbit overshoot is smaller than its final frame interval. CPU slowdown need not reduce refresh rate: this small scene still runs near 60 FPS. Gaps exceeding 0.1 s lose elapsed time under the clamp. The unclamped return includes the hidden interval in one dt. The five-second timer has native-input overhead, explaining the measured 5.1263 s. CPU throttling was restored afterward. These are Edge observations; no Chrome/Firefox test is claimed.

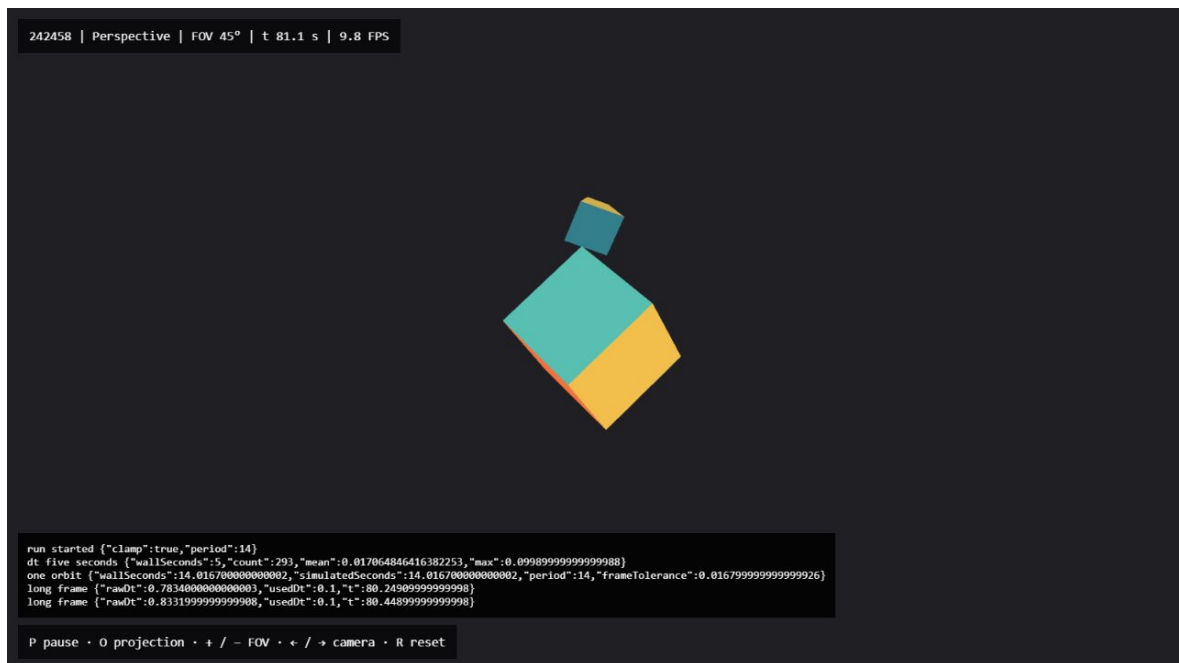


Figure 12. E5 unthrottled baseline and the first orbit results.

E5 - CPU throttling evidence

Native Edge DevTools Performance shows CPU 4x slowdown. The initial five-second sample and orbit finished before a later long suspension, also logged in this screenshot. That later suspension is excluded from the reported statistics.

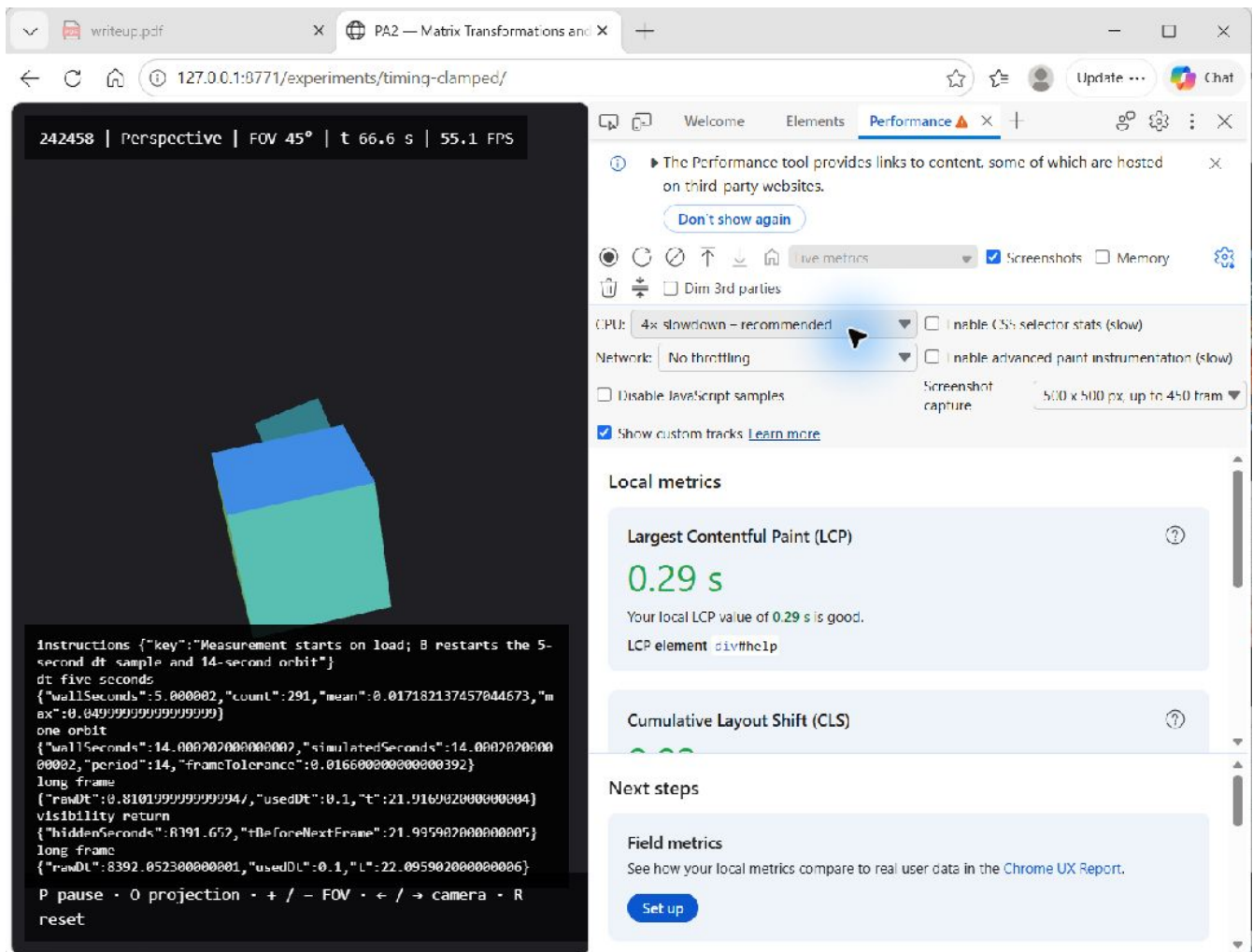


Figure 13. CPU 4x setting together with measured dt and orbit results.

E5 - Hidden-tab return without the clamp

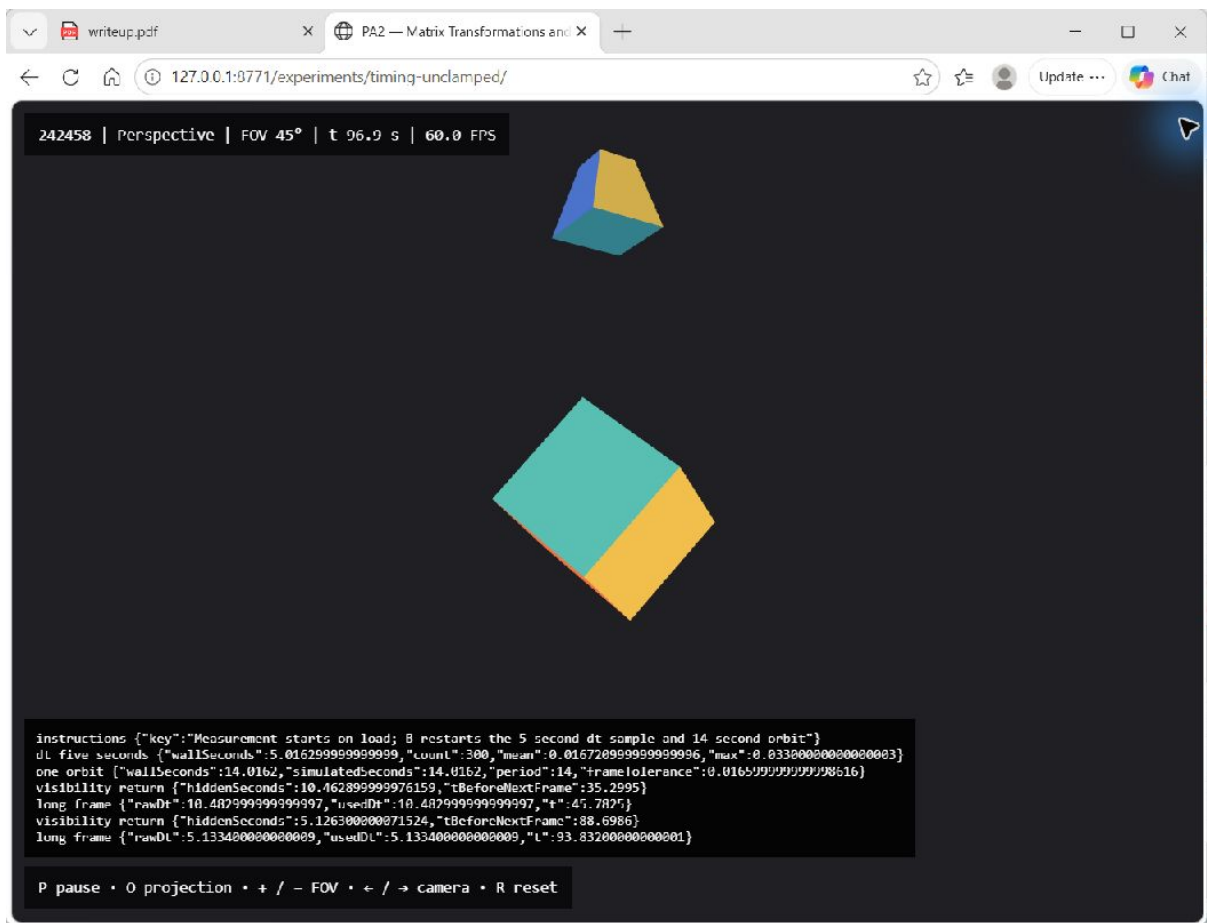


Figure 14. Actual hiding 5.1263 s; first returned dt 5.1334 s.

Native console logs show the same visibility and long-frame measurements. No errors appear in the captured console. CPU throttling had been restored before this test.

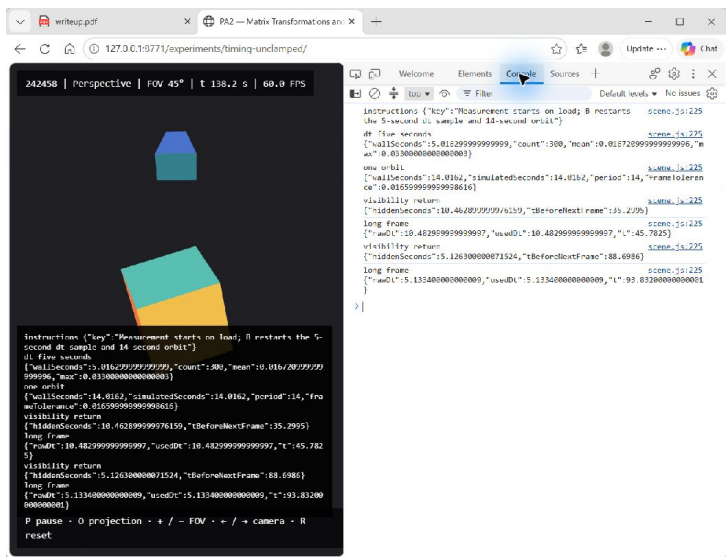


Figure 15. Console evidence of the unclamped return.

E6 - One vertex by hand

Prediction: $t=(2\pi/3)/1.2=1.745329252$ s produces 120° about $(1,1,1)/\sqrt{3}$. Rodrigues' formula simplifies to $M(x,y,z,w)=(z,x,y,w)$, so $v=(0.5,0.5,-0.5,1)$ maps to $(-0.5,0.5,0.5,1)$.

Result: `vec4.transformMat4` matches that world coordinate. Console clip coordinates are $(-0.6789975762, 0.7307890654, 6.6612720490, 6.7939953804)$. Dividing xyz by 6.7939953804 gives NDC $(-0.0999408357, 0.1075639627, 0.9804646126)$. Evidence: `vertex-by-hand.png` and `E6-output.txt`.

Explanation: the shader applies model, view and projection in that order. The final clip w, rather than the original vertex $w=1$, is the divisor. `glMatrix` prints column-major entries; tiny residuals near zero are rounding.

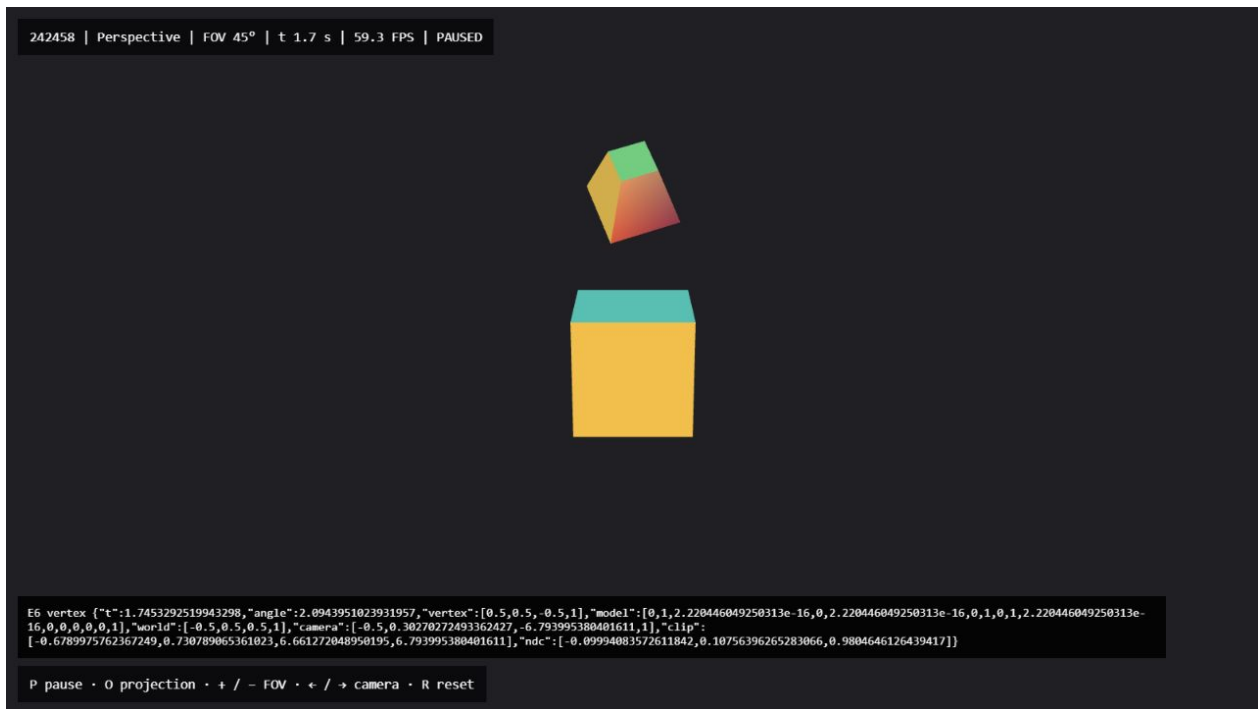


Figure 16. E6 at $t=1.745329252$ s with the world, clip and NDC console result.

Development log

The FPS counter initially counted the first timestamp without a frame interval. Comparing counted frames with timestamp intervals exposed the mismatch. Commit 390cba5 excludes the initial timestamp and measures windows of at least one second. Commit 0207e75 adds real CPU-throttling and hidden-tab evidence for E5.